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Enhancement of antibacterial activity by a copper(II) and zinc(II) in chelation with ethylenediaminetetra-acetic acid and urea complex

Trong Nghia Nguyen¹ · Quang Hoa Do¹ · Thi Thuy Duong Vu¹ · Thi Thuy Nguyen¹ · Duc Toan Nguyen¹ · Thi Bich Ngoc Nguyen¹ · Thi Thu Ha Tran¹ · Thi Kim Oanh Vu¹ · Thi Ha Lien Nghiem¹ · Thi Minh Tu Hoa² · Thi Minh Huyen Nguyen² · Phuong Nhung Bui³ · Minh Hue Nguyen⁴ · Duc Minh Pham⁵ · Thi Chung Thuy Tran⁶

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Abstract

Although it is unclear how Zn^{2+} cooperates with Cu^{2+} in synergistic antibacterial activity, a 1:10 ratio of $\text{Cu}^{2+}/\text{Zn}^{2+}$ atoms and ethylenediaminetetra-acetic acid and urea ligands can be used to form a chelation complex containing Cu^{2+} and Zn^{2+} . This study investigated the effects of the combination of Cu^{2+} and Zn^{2+} in chelation with EDTA and urea. The results were compared with the outcomes of either copper or zinc alone against gram-negative *Escherichia coli* (*E. coli*) and gram-positive *Staphylococcus aureus* (*S. aureus*) bacteria. The antibacterial activity was determined through MICs, disc diffusion method, and bacterial cell growth curves. In addition, bacterial destruction by this chelation complex has been observed through SEM images. The presence of copper ion and zinc ion inside the bacterial cells has been proved through EDS measurements. The obtained results allowed us to draw conclusions that the use of both Cu^{2+} and Zn^{2+} in a chelation complex with EDTA and urea enhances the antimicrobial activity against these bacteria. The bacterial inhibition of this complex was stronger than that of Cu^{2+} alone chelated with EDTA or CuSO_4 solution by approximately 100-fold for *S. aureus* and 20-fold for *E. coli*.

Keywords Antibacterial activity · Gram-positive bacteria · Gram-negative bacteria · Copper ion · Zinc chelate complex

Introduction

The use of antibiotics in aquaculture environment treatment poses a major threat to the health and well-being of humans. The discovery and development of antibiotics in the last

century have dramatically changed treatment outcomes for infectious diseases, including those in aquaculture environments (World Health Organization et al. 2006). However, the development of bacterial resistance against even the most powerful antibiotics is rapidly becoming a major challenge for the application of antibiotics in combating infectious diseases and in environmental treatment. Many studies have been focused on finding agents to substitute antibiotics in environmental treatment and in the treatment of bacterial infections in hospitals. Metallic ions have been emerging as substantial components (Stout et al. 1998). The antibacterial activity of metallic ions such as Cu^{2+} , Ag^+ , Ni^{2+} , and Co^{2+} was proven for individual ions (Xu et al. 2012) and is greatly increased in combination. Their antibacterial activity was also influenced by chelated molecules. The antibacterial and antifungal agents of Co^{2+} , Ni^{2+} and Zn^{2+} complexes with amino acid-derived compounds have also been evaluated by Chohan's research (Chohan et al. 2006). The research of Ibrahim and co-workers showed that one ZnCl_2 molecule could combine with two or four urea molecules ligand to form the two solid crystal urea complexes (Ibrahim et al. 2012). Recently, the improving mildew property of Bamboo

✉ Thi Ha Lien Nghiem
halien@iop.vast.vn

✉ Thi Minh Huyen Nguyen
ntminhhuyen@ibt.ac.vn

¹ Institute of Physics, VAST, 10 Daotian, Badminh, Hanoi, Vietnam

² Institute of Biotechnology, VAST, 18 Hoang Quoc Viet, Cau Giay, Hanoi, Vietnam

³ Faculty of Veterinary Medicine, Vietnam National University of Agriculture, Ha Noi, Vietnam

⁴ Department of Physics, Le Quy Don Technical University, Hanoi, Vietnam

⁵ 103 Hospital, Vietnam Military Medical University, 261 Phung Hung, Ha Dong, Hanoi, Vietnam

⁶ Thuyloi University, 175 Tayson, Dongda, Hanoi, Vietnam

of the nanometer zinc complex formed from ZnCl_2 -urea ion liquid has been proved by Gao. This study showed that using a 1:2 molar ratio of zinc chloride to urea, the complex possesses the strongest mildew resistance (Gao et al. 2020). From coordination chemistry, ethylenediaminetetra-acetic acid (EDTA) is a polyaminocarboxylate ligand that forms excellent chelators. There are two-electron donor groups: amino nitrogen and carboxyl oxygen. The amino nitrogen favourably attracts copper and zinc while the carboxylate function can chelate almost all metal ions. Its coordination sites are fully occupied by the same ligand molecule, having two such groups that can form all metal-EDTA complexes, which have 1:1 stoichiometry in one step (Jeffery et al. 1989). EDTA is the most frequently used chelating agent in land decontamination thanks to its ability to form a highly stable complex with most metallic ions. It enhances their solubility in aqueous solutions (Tsang et al. 2012). In medicine, its salt $\text{Na}_2\text{Ca}(\text{EDTA})$ is used for injection as an antidote to lead poisoning (Crichton et al. 2016). In an acidic medium, the copper (II) with EDTA forms a stable and intense blue Cu-EDTA complex (Jeffery et al. 1989). Recently, the synergistic mechanism of the combination $\text{Ag}^+/\text{Zn}^{2+}$ in antibacterial activity against *Enterococcus faecalis* and use against dentin infection was reported by Wei Fan. This study found that the concentration of silver ion needed for bacterial inhibition was approximately 3.10×10^{-2} mM (Fan et al. 2018). $\text{Ag}^+/\text{Zn}^{2+}$ atomic ratios of 1:9 and 1:12 demonstrated a similar strong ability against *Enterococcus faecalis* biofilms on dentin; moreover, their toxicity was lower than that of 2% chlorhexidine (Fan et al. 2018, a). We also know that zinc and copper are essential trace elements for many biological functions. They play crucial roles in the organism's growth, cell division, protein synthesis, and the activity of the metalloenzymes. Cu^{2+} and Zn^{2+} ions have roles in host cell growth inhibition, and Zn^{2+} ion inhibits viral entry, local replication, and spread to organs during the viral pathogenesis process (Ishida et al. 2019; Ishida et al. 2018; Soliman et al. 2020). In humans, the daily intake recommendations for zinc and copper for adult persons are 15 mg and 0.9 mg, respectively (Institute of Medicine (US) and Panel on Micronutrients. 2002; Al-Fartusie et al. 2017). Nevertheless, copper and zinc ions are very toxic to unicellular organisms and mammalian cells in vitro. Thus, the antibacterial activity of these ions and their oxide nanoparticles has been previously researched (Ishida et al. 2019; Borovansky et al. 1989; Diaz-Visurraga et al. 2011; Solioz et al. 2018; Ishida et al. 2017; Gopalakrishnan et al. 2009). The antibacterial activities of Cu^{2+} ions against *E. coli* and *S. aureus* were intensively researched. The results showed minimum inhibitory concentration (MIC) values of 625 mg/L against *S. aureus* (Ishida et al. 2017). Many researchers have investigated the combination of Zn^{2+} ion or Cu^{2+} ion with the chelate agents such as hydrazone ligands to enhance their antibacterial and

antifungal. However, the antibacterial/antifungal ability of these complexes was studied at the individual metals either Zn or Cu complexes (Santiago 2020; Cholan 2006; Paladini 2015; Rizzotto 2012; Yermale 2020). To date, there are few reports about the bactericidal activity of the combination of Cu^{2+} and Zn^{2+} compared with each metal ion alone. Copper-zinc hybrid nanoparticles were used to treat bacterial spots in tomatoes caused by copper-tolerant bacteria (Carvalho et al. 2019). However, the paper did not emphasize the effect of the combination in comparison with either copper or zinc alone. The antibacterial and antifungal activities of montmorillonites (MMT) containing silver, copper, or zinc were also described by Malachova (Malachova et al. 2011). In their paper, different bacteria were inhibited differently by each MMT. Jiao et al. showed that Cu/Zn MMT had enhanced the antibacterial activity compared with those of Zn-MMT and Cu-MMT in their work (Jiao et al. 2017). However, the Cu/Zn loading on the solid-state MMT mineral by the ion exchange process was complicated and the ratio of Cu/Zn with the highest effect was 0.98:0.51. The antibacterial activity effects of Cu/Zn-MMT are caused by these ions with unknown concentrations due to their slow-releasing process into the media.

This work aims to find an easy method to make high-efficiency antibacterial solutions containing these two essential trace elements Cu^{2+} and Zn^{2+} ions using the two environmentally friendly chelating agents such as EDTA and urea. Therefore, we investigated the effects of combining Cu^{2+} and Zn^{2+} chelated with EDTA and urea in an aqueous solution and compared the results with the antibacterial effect against gram-negative *Escherichia coli* (*E. coli*) and gram-positive *Staphylococcus aureus* (*S. aureus*) bacteria of either copper or zinc. The special point of our experiment is that the Cu/Zn ratio of 0.1 was well controlled and optimized to be closer to the ratio of $\text{Cu}^{2+}/\text{Zn}^{2+}$ (0.9:15 or 0.06) required for human health (Al-Fartusie et al. 2017; Zoroddu et al. 2019). The antibacterial activity was determined through MICs, disc diffusion method, and bacterial cell growth curves. Nevertheless, our results obtained in the disc diffusion assay for both types of bacteria showed that the bactericidal effect of Cu^{2+} chelate was higher than that of Zn^{2+} chelate but much lower than the combination of $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate. The obtained results allowed us to conclude that the use of both Cu^{2+} and Zn^{2+} in a chelation complex with EDTA and urea enhances the antimicrobial activity against these bacteria. The bacterial inhibition of this complex was stronger than that of Cu^{2+} alone chelated with EDTA or CuSO_4 solution by approximately 100-fold for *S. aureus* and 20-fold for *E. coli*.

Materials and methods

Reagents

Urea ($\text{N}_2\text{H}_4\text{CO}$) and disodium ethylenediaminetetraacetate dehydrate ($\text{C}_{10}\text{H}_{14}\text{N}_2\text{Na}_2\text{O}_8$, EDTA.2Na.2H₂O), $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ and ZnCl_2 were obtained from Xilong Scientific. Agar origin in Vietnam. Sterile Whatman No. 1 filter paper was purchased from Sigma-Aldrich. Gram-negative *E. coli* strain ATCC11105 and Gram-positive *S. aureus* strain ATCC13709 bacteria were obtained from the collection of the Institute of Biotechnology, VAST. The bacterial strains were cultured in Luria Bertani (LB) broth or agar plates (peptone 10 g/L, yeast extract 5 g/L, NaCl 10 g/L, pH ~ 7.0; agar 15 g/L). Distilled water was used in the preparation of the chelate complexes.

Preparation of the zinc (II) and copper (II) chelate complexes

According to the previous research (Ibrahim et al. 2012; Gao et al. 2020), and the requirement of two metals for human health (Al-Fartusie et al. 2017; Zoroddu et al. 2019), we prepare a mixture of two complexes Cu-EDTA and $\text{ZnCl}_2 \cdot 4\text{Urea}$, and in order to obtain a solution, which contains the two above metal complexes and have the Cu/Zn molar ratio of 0.1. The chelate complexes were prepared by dissolving $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (0.05 M) and ZnCl_2 (0.5 M) with urea (1 M) and EDTA.2Na 2H₂O (0.05 M) in aqueous solutions. After thorough mixing of salts and ligands by ultrasonication, stock solutions of zinc(II) and copper(II) chelate complex were obtained. The zinc (II) and copper (II) chelate complexes and single metal complexes from these precursors were prepared in an aqueous solution with the corresponding concentrations. These complexes are considered stock solutions (G), which were used for experiences without purification. The characterizations of all these solutions including the zinc (II) and copper (II) chelate complexes and their single complex in aqueous solutions were obtained by measuring their UV–VIS–NIR spectra on a Shimadzu 2600 spectrophotometer.

Determination of MICs

Minimum inhibitory concentrations (MICs) were determined by incubating the bacterium in a serial dilution of 2 times of ion solution (Wiegand et al. 2008). The optical density at 600 nm of the original culture of bacteria was adjusted to approximately 0.1. The bacterial solution was incubated overnight in a shaking incubator at 37 °C and 200 rpm. The optical density values at 600 nm of the overnight bacterial

cultures were determined by a spectrometer. The minimal concentration of ion solution that inhibited the growth of bacteria was determined to be the MIC. The control experiment was the bacteria incubated without ion solution (also considered as zero molar concentration). The experiments were repeated 3 times. The value was the average of each measurement.

SEM image observation and EDS measurement

Scanning electron microscopy (SEM) images of *S. aureus* bacteria after 4 h and 24 h of incubation with single ions diluted 100-fold from the stored solutions and their prepared combination between Cu^{2+} and Zn^{2+} at dilutions of 10-, 100-, and 1000-fold from the stored combination solutions. After incubation with these agents, the bacteria were treated with the protocol A of Ammar (Ammar O 2017). Briefly, the bacterial broth was centrifuged to remove the excess ions. Then, the pellet was washed with phosphate buffer saline (PBS) 1× before fixing the bacteria with 3% paraformaldehyde solution for 1 h. After that, the pellet was washed 1 time with PBS 1x. These bacterial samples were dehydrated by incubation for 10 min with each of several different ethanol volumes: 30%, 50%, 70%, 80%, 90%, and 100%. Finally, each bacterial sample in ethanol was doped on silicon substrates and sputtered with a titanium layer for 20 s for observation by a Hitachi scanning electron microscope (Regulus 8100). The main components at different positions, inside and outside of bacteria were analysed by electron-dispersive X-ray spectroscopy (EDS) measurements at a high voltage of 30 keV.

Bacterial cell growth curve

The antibacterial activity of metal ions was also tested by measuring the optical density at 600 nm during the growth period of bacteria with various concentrations of metal ions. The details of the experiments are as follows: Cells were grown from 2% overnight seed culture in LB medium at 37 °C in a shaking incubator at 150 rpm. Different concentrations of metal ions were added. The growth rates and bacterial concentrations were determined by measuring the optical density at 600 nm every 60 min and drawing graphs. The experiments were performed in triplicate.

Disc diffusion assay

The disc diffusion method was used to test the effects of the metal ions at several concentrations on the survival of gram-negative and gram-positive bacteria, following the Kirby–Bauer disc diffusion susceptibility test protocol with slight modification (Hudzik 2009; Wiegand et al. 2008). The experiment was performed as follows: The agar plates

were prepared by autoclaving LB agar at 121 °C for 15 min and poured into 10-cm-diameter Petri dishes at 20 ml/plate. The agar was allowed to solidify, and the whole surface of the agar was covered with an overnight culture of bacteria by adding 1.5 ml of bacterial liquid culture that covered the whole surface of the plate. The excess bacterial liquid was removed, and the surface of the plate was allowed to dry completely. On the surface of the agar, a sterile Whatman No. 1 filter paper disc (6 mm diameter) loaded with 20 µl of a serially diluted ion solution (Table 1) was placed. In the control experiments, a paper disc loaded with 20 µl of autoclaved distilled water (DW) was used. The plates were incubated at 37 °C overnight. Each sample was repeated with 3 plates at each time point in 3 experiments. The clear zone (zone of inhibition, ZOI) surrounding the paper disc was determined by measuring the diameter of that zone. The results are the mean value of all measurements.

Results and discussion

The UV–VIS–NIR characterizations of zinc (II) and copper (II) chelate complexes

The UV–VIS–NIR spectra of the ion Zn^{2+} and Cu^{2+} in aqueous solution and all their complexes with EDTA and Urea are shown in Fig. 1. A, B, C. Figure 1. D is the absorption spectra of Fig. 1. C in energy form.

Figure 1 presents the absorption spectra of EDTA, urea, and their mixtures in aqueous solutions at the concentrations used in the chelate complexes. These spectra show that these agents are well soluted in DI water and they do not absorb light in the range of 250 nm to 1200 nm. Figure 1. B presents the absorbance spectra of ZnCl_2 and its complex with EDTA and urea. ZnCl_2 does not dissolve well in DI water and in urea solution, these solutions become turbid and have strong light absorption in the whole spectral range of the spectrometer. This is because of the hydrolysis of ZnCl_2 in water to form $\text{Zn}(\text{OH})_2$. These solutions become transparent after adding 0.05 M EDTA. $2\text{Na}_2\text{H}_2\text{O}$. These results show that it is easy to form the Zn-EDTA complex from ZnCl_2 and EDTA. Figure 1. C shows the absorbance

spectra of CuSO_4 , EDTA urea in aqueous solutions. The absorption spectra of CuSO_4 of 0.05 M in DI water and with urea 2 M have nearly the same shapes and intensity with a wide band and peak at $12,500\text{ cm}^{-1}$ (800 nm). The broadness of this band is also usually observed in simple octahedral copper (II) complexes. This band becomes intensive after the formation of the complex with EDTA, their absorbance increases 4 times from 0.6 to 2.30 compared to CuSO_4 solution at 0.05 M and 2.44 for the mixture of CuSO_4 and urea solutions. These absorption peaks also simultaneously shifted to $13,450\text{ cm}^{-1}$ (748–750 nm). This band is characteristic of ${}^2\text{E}_g$ ${}^2\text{T}_{2g}$ “d-d” transition of Cu^{2+} cation in an octahedral environment, which agrees with the previous study (Lakshmi et al. 2012). The reasons why the absorption band is shifted to higher energy and becomes sharper, and more intensive are well documented in Qui’s work (Qiu et al. 2015). Briefly, it is due to the process where EDTA ligands displace the water molecules to form a new mixed metal–ligand complex. The hybridization of the Cu^{2+} d electrons will perturb the electronic orbitals and reduce the symmetry leading to a higher probability of d–d transition. The absorption band at $35,000\text{ cm}^{-1}$ (300 nm), and the strong high energy band were shifted to the lower energy band at $25,000\text{ cm}^{-1}$ (400 nm). The shift of this strong high energy is assigned to ligand metal charge transfer. In addition, a new band appeared via the long wavelength at 8800 cm^{-1} , which is slightly decreased with the presentation of the urea and then ZnCl_2 agents. With the addition of urea to the Cu-EDTA complex, the absorbance of the spectral band at 750 nm is slightly increased. These results indicate that urea forms the weak complex with Cu^{2+} and Zn^{2+} , and also showed that Zn^{2+} ion hardly forms the Zn-EDTA complex in the presence of Cu^{2+} ion in solution. The results can be explained that the Cu-EDTA complex has higher stability constant of 18.8; hence, it is more stable than ion Zn-EDTA whose stability constant is 16.7 (Jeffery et al. 1989). This result also suggests that in the Zn^{2+} and Cu^{2+} chelate complexes solution formation from CuSO_4 , ZnCl_2 , EDTA, and urea agents, the Cu^{2+} ion is mainly associated with EDTA in the form Cu-EDTA complex, which also implies that Zn^{2+} ion has a weak connection with urea molecule

Table 1 The concentrations of the copper ion solutions used in the disc diffusion assay

Symbol on agar plates	CuSO_4 (mM)	Cu^{2+} chelate (mM)	$\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate (mM)	Zn^{2+} chelate (mM)	EDTA (mM)
(G)	100	50	–	500	50
(– 1)	10	5	5/50	50	5
(– 2)	1	0.5	0.5/5	5	0.5
(– 3)	0.1	0.05	0.05/0.5	0.5	0.05
(– 4)	–	–	0.005/0.05	–	–
(DC)	DW	DW	DW	DW	DW

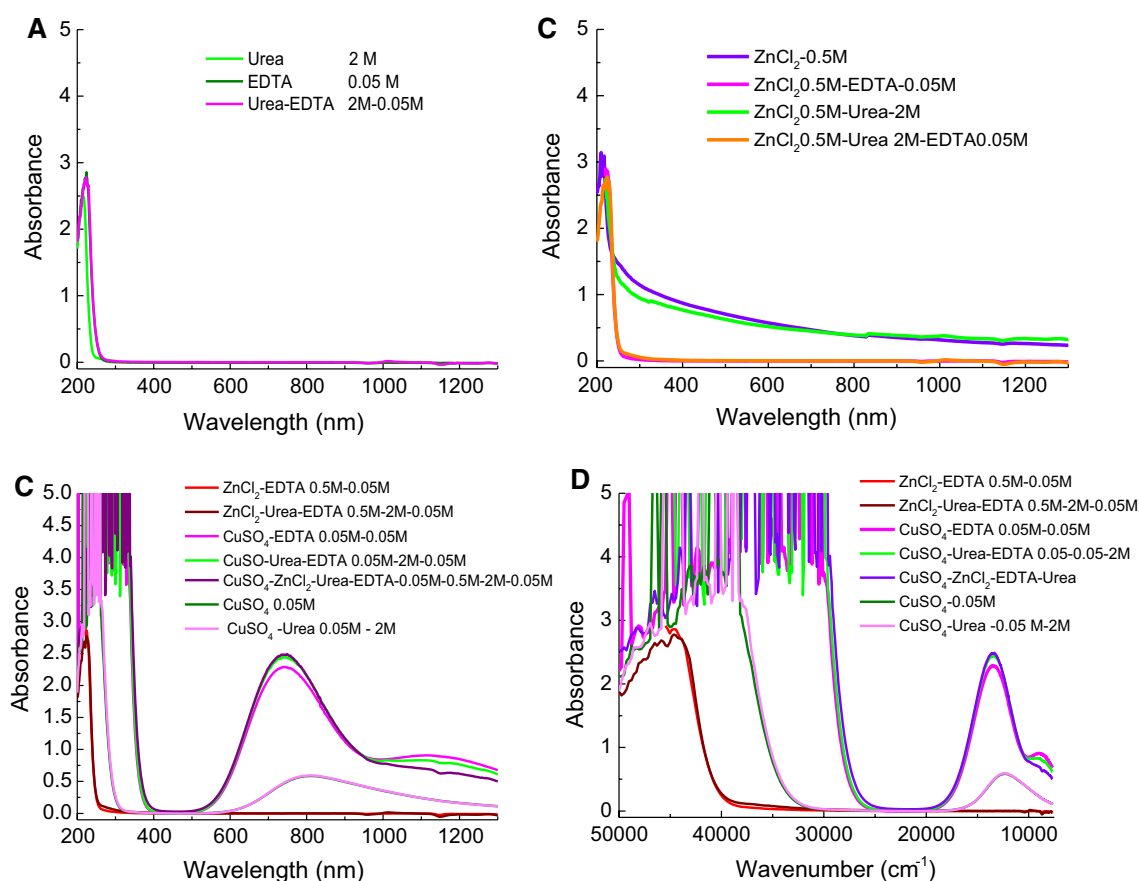


Fig. 1 UV–VIS–NIR spectra of the metal ions and chelate complexes

MIC determination

The MICs of CuSO_4 and Cu^{2+} chelate (Fig. 2) in both *E. coli* and *S. aureus* were 5 mM and 10 mM, respectively. This concentration was 5 and 20 times higher than the highest tested concentration in the bacterial cell growth experiment. Therefore, the results of the bacterial cell growth curve were consistent with the MIC results with CuSO_4 and Cu^{2+} chelate. With Zn^{2+} chelate and $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate complexes, there were differences in MIC values between gram-negative bacteria and gram-positive bacteria. For *E. coli* bacteria, the MIC values were 3.125 mM and 0.625 mM/6.25 mM for the Zn^{2+} chelate and $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate complexes, respectively. For *S. aureus* bacteria, the MICs were at least 2 times lower (1.5625 mM and 0.15625 mM/1.5625 mM for Zn^{2+} chelate and $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate complexes, respectively). The MICs of the CuSO_4 solution were twofold lower than those of the Cu^{2+} chelate solution, which is consistent with the MIC value found by Ishida T. of approximately 625 ppm (Ishida et al. 2017). This result can be explained by the free ion of Cu^{2+} being more active than its EDTA chelating complex. Moreover, the MIC values of these solutions against

gram-positive bacteria were twofold lower than those against gram-negative bacteria.

Mechanism of the antibacterial activity of Cu^{2+} and Zn^{2+} ions

The mechanism of the antibacterial activity of Cu^{2+} ions on gram-negative and gram-positive bacteria has been detailed in Marc Solioz's book and Ishida's research (Solioz 2018; Ishida et al. 2017). This result can be explained by the different cell wall components of gram-negative and gram-positive bacteria. Gram-positive bacteria have a thick peptidoglycan layer and no outer lipid membrane, while gram-negative bacteria have a thin peptidoglycan layer and an outer lipid membrane. The bacteriolytic mechanism of Cu^{2+} ions against the *E. coli* cell wall proceeds by the destruction of the outer membrane structure by the degradation of outer membrane lipoproteins at the N- and C-termini and the inhibition of peptidoglycan elongation due to the deletions of the peptidoglycan transpeptidase enzyme and peptidoglycan autolysins. The other bacteriolytic mechanism of Cu^{2+} ions against the *S. aureus* cell wall consists of damage to the peptidoglycan synthesis enzyme transglycosylase due to the polymerization of glycan

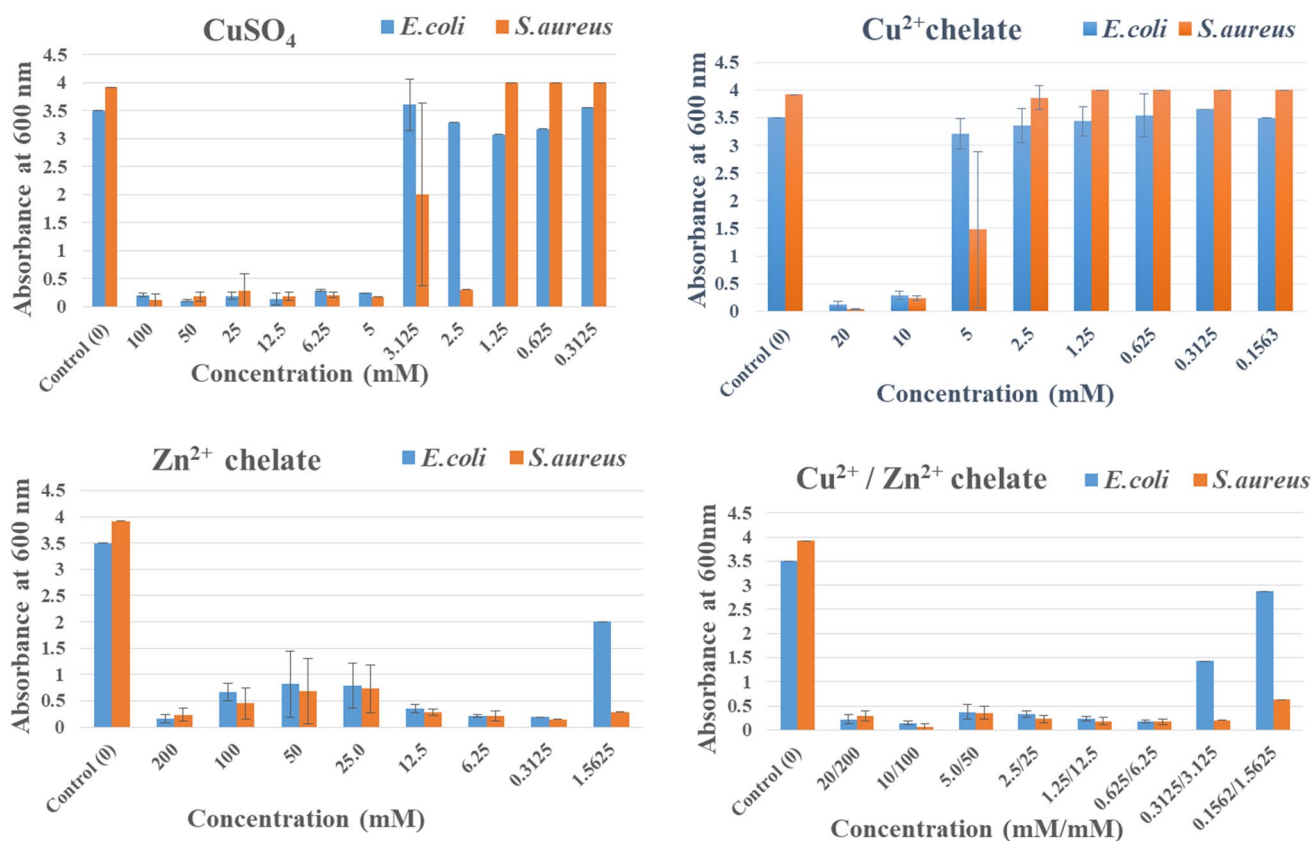
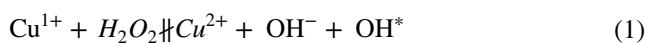


Fig. 2 MICs of metal ions and chelating agents, controls are bacteria without treatment

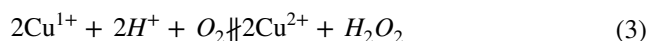
chains, the cross-linking of peptide chains, and the activation of peptidoglycan by autolysins (Ishida et al. 2017).

After penetrating into bacteria, copper ions react with H_2O_2 in cells to produce OH^* free radicals (highly reactive free hydroxyl radicals).



These highly active hydroxyl radicals can participate in several adverse reactions to cellular molecules, such as the oxidation of proteins and lipids, resulting in the destruction of living organisms with simple structures, such as bacteria.

In addition, in cells, copper ions can combine with sulfhydryl molecules such as glutamine/cysteine, disrupting metabolic mechanisms and leading to the destruction of the cell, as shown in Eq. (2). The recreated Cu^{2+} ion can combine with the oxygen inside the cell to produce H_2O_2 molecules as a source for creating hydroxyl free radicals with high reactivity. Therefore, copper ions can also be created by the depletion of sulfhydryl, such as cysteine or glutathione molecules, in a cycle between reactions (2) and (3):



Therefore, the generated hydrogen peroxide can, in turn, participate in reaction (1) and lead to the generation of additional free-radical toxic hydroxyl groups. These hydroxyl radicals continue to exhibit strong activity in destroying the cells of simple living organisms, as shown in Eq. (1) (Ishida et al. 2017; Solioz 2018).

Research by Solioz has also shown that the sulfhydryl molecules inside microorganisms form complexes with copper ions via Eq. (4).



Under anaerobic conditions, the sulphuryl copper complex can act as a copper donor for metalloenzymes. The dominant toxicity mechanism is most likely the displacement of iron from iron-sulphur cluster proteins by Cu^{1+} , rendering key enzymes inactive (Solioz 2018). The different structures and components on the cell walls of gram-negative and gram-positive bacteria caused the different MIC values of the two bacterial types (Ishida et al. 2017). The research of Solioz on copper toxicity showed that the Zn^{2+} ion has the same antibacterial mechanism as the Cu^{2+} ion, with lower

toxicity. However, in this experiment, we observed Zn^{2+} ion to be as toxic as or even more toxic than Cu^{2+} ion. The MIC values of the chelated Zn^{2+} were 1.562 mM and 3.125 mM for gram-negative and gram-positive bacteria, respectively, which are three times smaller than those of the Cu^{2+} chelate solution. Only free copper ion or free zinc ion is biologically active, but the concentration of these free ions is unknown in their experiments (Solioz 2018). This result suggested that in nutritionally rich LB medium, the concentration of free Cu^{2+} ion is lower than that of free Zn^{2+} ion, or the chelating or binding between amino acids or/and proteins via amino or thiol groups with Cu^{2+} ion is stronger than with Zn^{2+} ion.

SEM and EDS results

Figure 3 presents SEM images of *S. aureus* bacteria after 4 h and 24 h of incubation with PBS at room temperature without shaking. No slimy extracellular material was observed, which suggested it may have been dissolved or washed away. We observed aggregation of bacteria, which may be due to the beginning of microcolony formation. Over 24 h, long thin extracellular fibres were produced for attachment to each other, indicating biofilm formation (Barnes et al. 2012).

Figure 4 presents SEM images of the *S. aureus* incubated for 4 h and 24 h with single ion solutions and the combination of Cu^{2+} and Zn^{2+} at different dilution factors. The corresponding concentrations are shown in Table 1 and Fig. 4. SEM imaging showed that after 4 h of incubation (Fig. 4. A), only the complex of $\text{Cu}^{2+}/\text{Zn}^{2+}$ at the highest concentration (5 mM/50 mM) destroyed the *S. aureus* bacteria. All other *S. aureus* samples, including *S. aureus* incubated with Zn^{2+} chelate at 5 mM and the $\text{Cu}^{2+}/\text{Zn}^{2+}$ complex at 0.5 mM/5 mM, showed similar bacteria survival to those incubated with PBS 1x. The results showed greater destruction of *S. aureus* bacteria after 24 h of incubation with combined $\text{Cu}^{2+}/\text{Zn}^{2+}$ complex at 0.5 mM/5 mM, as shown in

Fig. 4. B. SEM images showed that the *S. aureus* bacteria incubated with single ions, i.e. with Cu^{2+} chelate and CuSO_4 (0.5 mM) and with Zn^{2+} chelate (5 mM), looked similar to those at 4 h. The obtained results after 24 h of incubation matched the MIC values found above, except for the *S. aureus* bacteria incubated with the Zn^{2+} chelate (5 mM). This result shows that at the MIC, these ion solutions inhibited bacterial growth but did not kill the bacteria. Adding 0.5 mM Cu^{2+} ions into the Zn^{2+} chelate solution 5 mM or using the combination of $\text{Cu}^{2+}/\text{Zn}^{2+}$ (0.5 mM/5 mM) successfully damaged or destroyed *S. aureus* bacteria.

Figure 5 presents SEM images of *E. coli* bacteria incubated for 4 h with these ion solutions, and the concentrations used were close to the MIC values, except in the case of Zn^{2+} chelate, which was 50 mM (15-fold higher than its MIC value). The three concentrations of the combined $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate complex were the same as those applied to *S. aureus* bacteria. The SEM images showed that the *E. coli* bacteria were destroyed in all samples, except at the lowest concentration of the $\text{Cu}^{2+}/\text{Zn}^{2+}$ complex, for which the *E. coli* had not yet been destroyed. These SEM images also suggest that at the MIC values of copper ion and the copper and zinc ion complex, these solutions are able to inhibit both bacterial growth and biofilm formation.

SEM images of *E. coli* bacteria before (Fig. 6.A) and after incubation with the combined $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelation complex at 0.5 mM/5 mM $\text{Cu}^{2+}/\text{Zn}^{2+}$ ions for 4 h (Fig. 5.B). The *E. coli* bacterial control sample has a plate ring, which could be the outer membrane proteins. We do not clearly see damage to the cell walls or the appearance of pits/cavities on the bacteria after four hours of incubation with the $\text{Cu}^{2+}/\text{Zn}^{2+}$ complex (Fig. 6B). However, SEM images showed that the materials around these bacteria became brighter. This implied that the Zn^{2+} ion and Cu^{2+} ion bound with the outer walls of the bacteria (Fig. 6.B). The bright materials had nanosized and spongy nanoporous structures, resembling

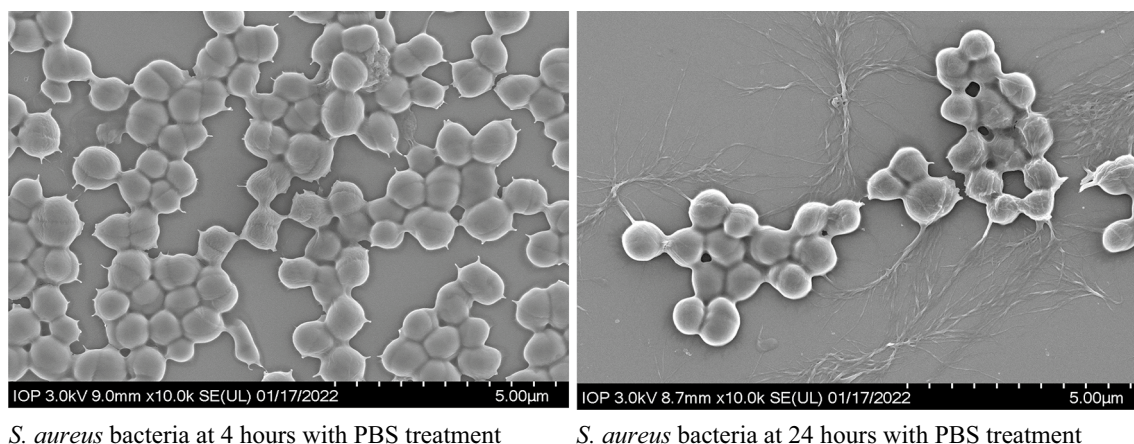


Fig. 3 SEM images of *S. aureus* after 4 h and 24 h of incubation with 1×PBS

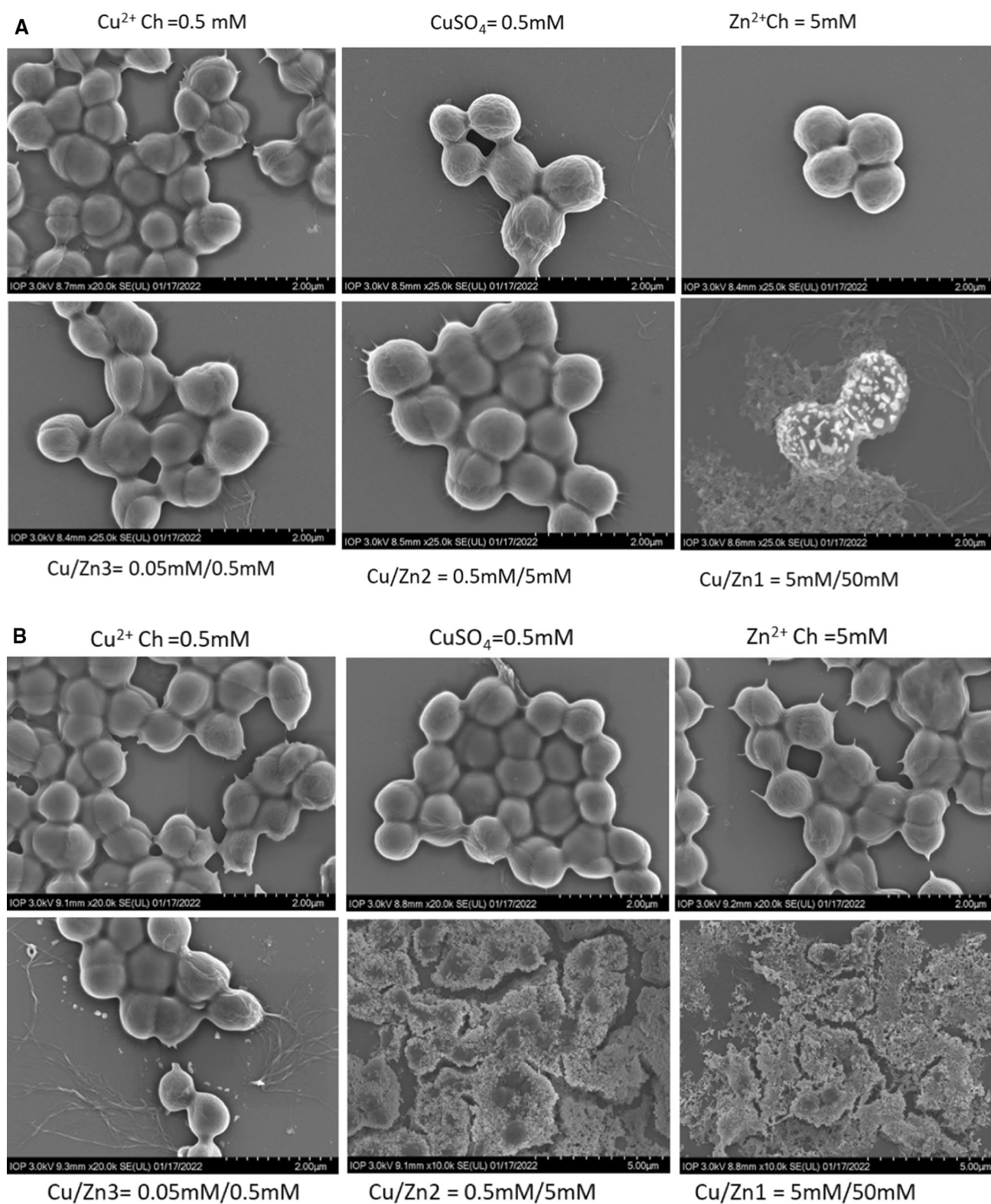


Fig. 4 SEM images of *S. aureus* incubated with single-ion solutions and a combination of Cu^{2+} and Zn^{2+} ions at different dilutions for 4 h (A) and 24 h (B)

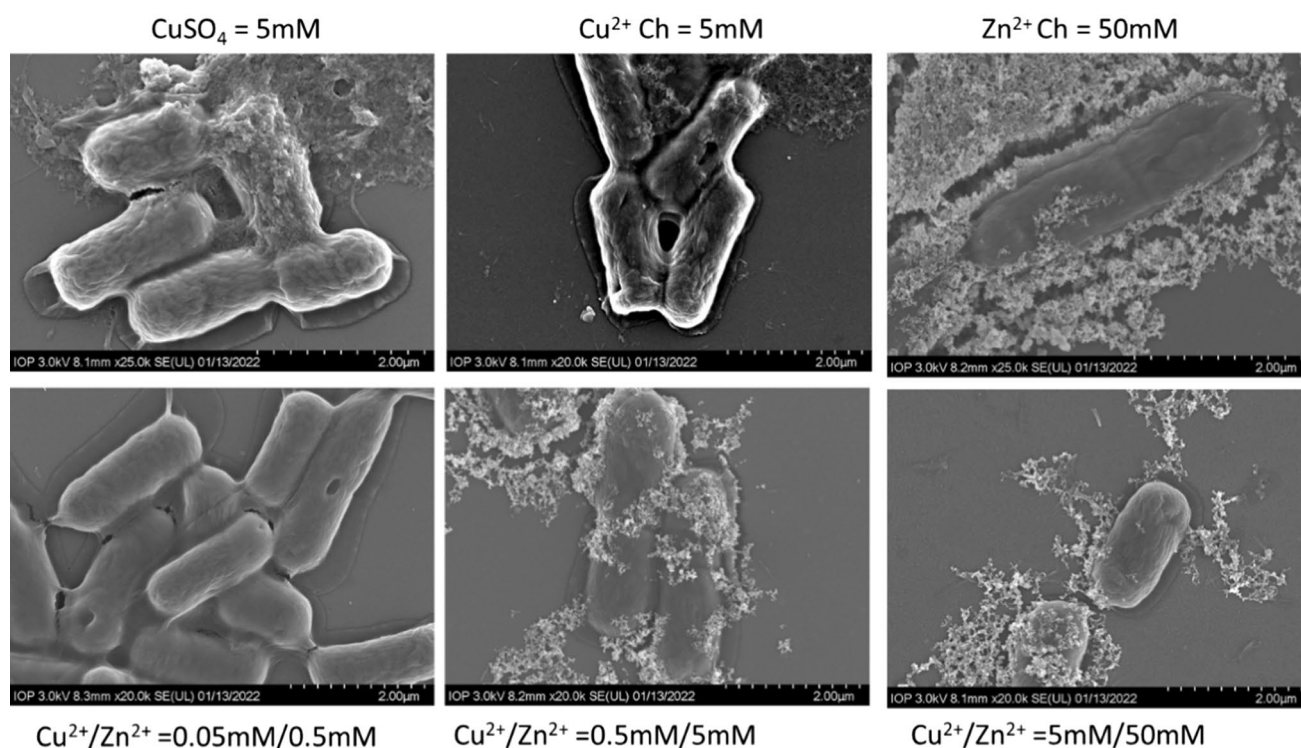


Fig. 5 SEM images of *E. coli* incubated with single ion solutions and a combination of Cu^{2+} and Zn^{2+} ions at different dilutions after 4 h of incubation



Fig. 6 SEM images of *E. coli* before **A** and after **B** incubation with the combination of $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate at 0.5 mM/5 mM $\text{Cu}^{2+}/\text{Zn}^{2+}$; and **C** schematic illustration of the various hypothesized interactions between metal ions and bacterial cells (Diaz-Visurraga et al. 2018)

metallic oxide nanoparticles. It is possible that ZnO or $\text{CuO}/\text{Cu}_2\text{O}$ nanoparticles formed from Cu^{2+} and Zn^{2+} ions in the LB medium. These observed phenomena are consistent with the hypothesized interactions between the bacteria and metal ion solutions, which have also been detailed in the work of the A. Garcia group, as shown in Schematic 6.C, which illustrated the probable mechanisms involved in metal ion and nanostructure activity on bacteria including: (1) metal ion uptake into cells, followed by depletion of intracellular ATP production and disruption of DNA replication; (2) release of metal ions and generation of reactive oxygen species; and (3)

accumulation and dissolution of metal nanoparticles in the bacterial membrane (Diaz-Visurraga et al. 2018).

The main components at four different positions of interest were analysed by EDS measurements at a high voltage of 30 keV, as shown in Fig. 7. The bacterial samples for EDS measurement were not sputtered. The results showed that carbon (C), nitrogen (N), and oxygen (O) dominated the mass. On the outer bacterial wall, C, N, and O accounted for approximately 88% to 91% of mass, while zinc was approximately 8% to 12%. In particular, the mass of copper was zero percent. This result again proved that in an LB solution,

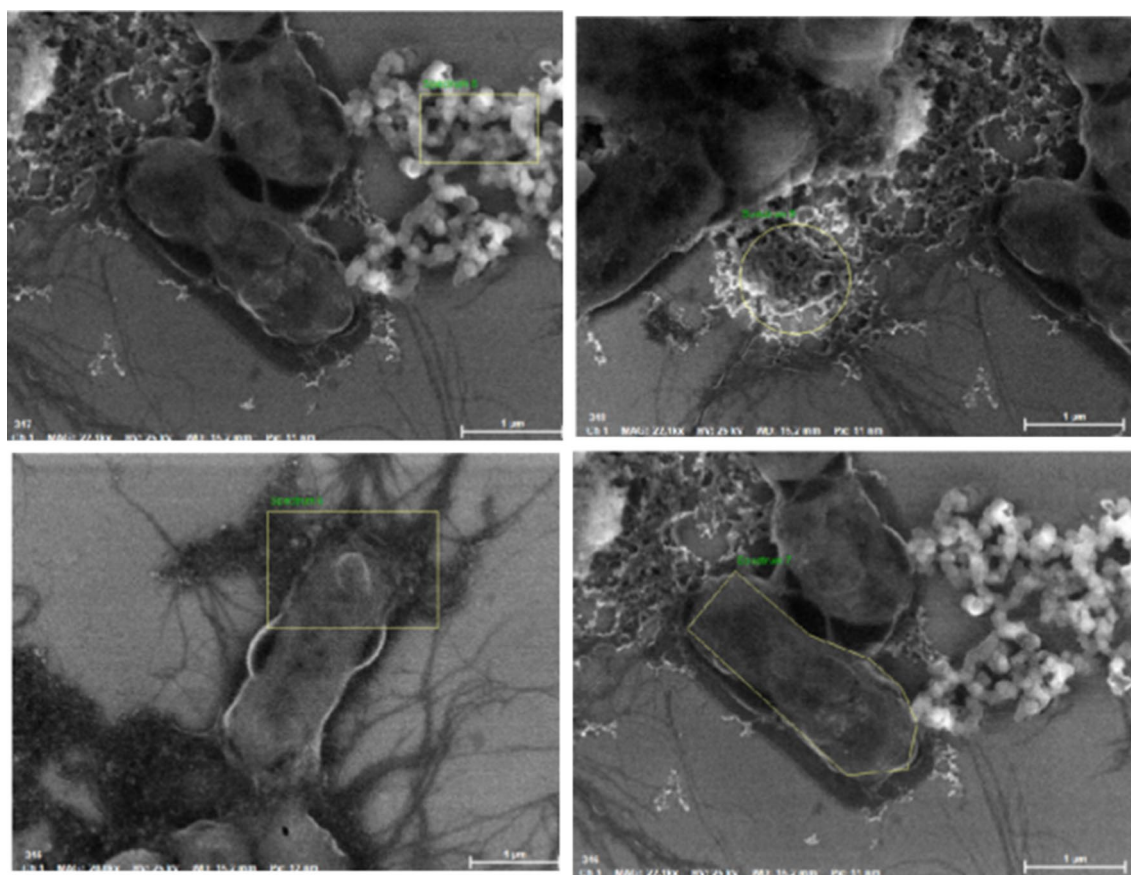


Fig. 7 EDS of *E. coli* incubated with the $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelation complex at different positions, outside and on the bacterial body

abundant zinc ions form a complex with the amino groups of proteins in the outer bacterial walls. Inside the bacteria, the EDS results showed the presence of calcium, sodium, and chloride ions in the bacterial cytoplasm. More specifically, the EDS results showed both copper and zinc metals with copper representing 0.26% to 0.61% of the mass and zinc representing 1.16% to 2.59%. These results proved that copper and zinc ions penetrated inside the bacteria. These results can be explained that in the metal-EDTA complex, the positive charge of the metal ion is partially shared with the hetero electron donor atoms (N and O) of the EDTA ligand and there may be π -electron delocalization over the whole chelating system, as a result, the lipophilic character of the metal chelates was increased and favour their permeation through the lipid layer of the bacterial cell membranes and blocking of the metal-binding sites in the metalloenzymes for superoxide of dismutase of microorganisms (Mau-rya and Namdeo 2021, Sharma 2018). These complexes also disturb the respiration process of the cell and thus block the synthesis of protein that restricts further growth of the organism. This result confirms that the combination of the zinc(II) and copper(II) complex with EDTA enhanced the antibacterial activity (Rizzotto 2012; Chohan 2006).

Effect of metal ions on bacterial cell growth in liquid culture

Bacterial cell growth under different concentrations of metallic ions was also tested in liquid media. The concentration of metal ions in the solution is the final concentration in each bacterial cell solution. The cell growth curves were plotted from data collected every hour (Fig. 8). With EDTA at the concentration tested in the experiment, there was no inhibitory effect on bacterial cell growth (Fig. 8 A.1, A.2). According to Fig. 8.C and 8.D, CuSO_4 and Cu^{2+} chelate had a slight inhibitory effect on both types of bacteria at the highest tested concentrations of 1 mM and 0.5 mM, respectively. Other lower concentrations of these chemicals had no difference in effect from the control experiment. Therefore, the chelating agent at the tested concentration did not play any role in reducing the cell growth rate, and the inhibitory effect came from the metal ions themselves. The Zn^{2+} chelate itself had a strong inhibitory effect at the highest concentration of 5 mM in both types of bacteria, in which cell growth was completely suppressed after 7 h of the experiment. However, at lower

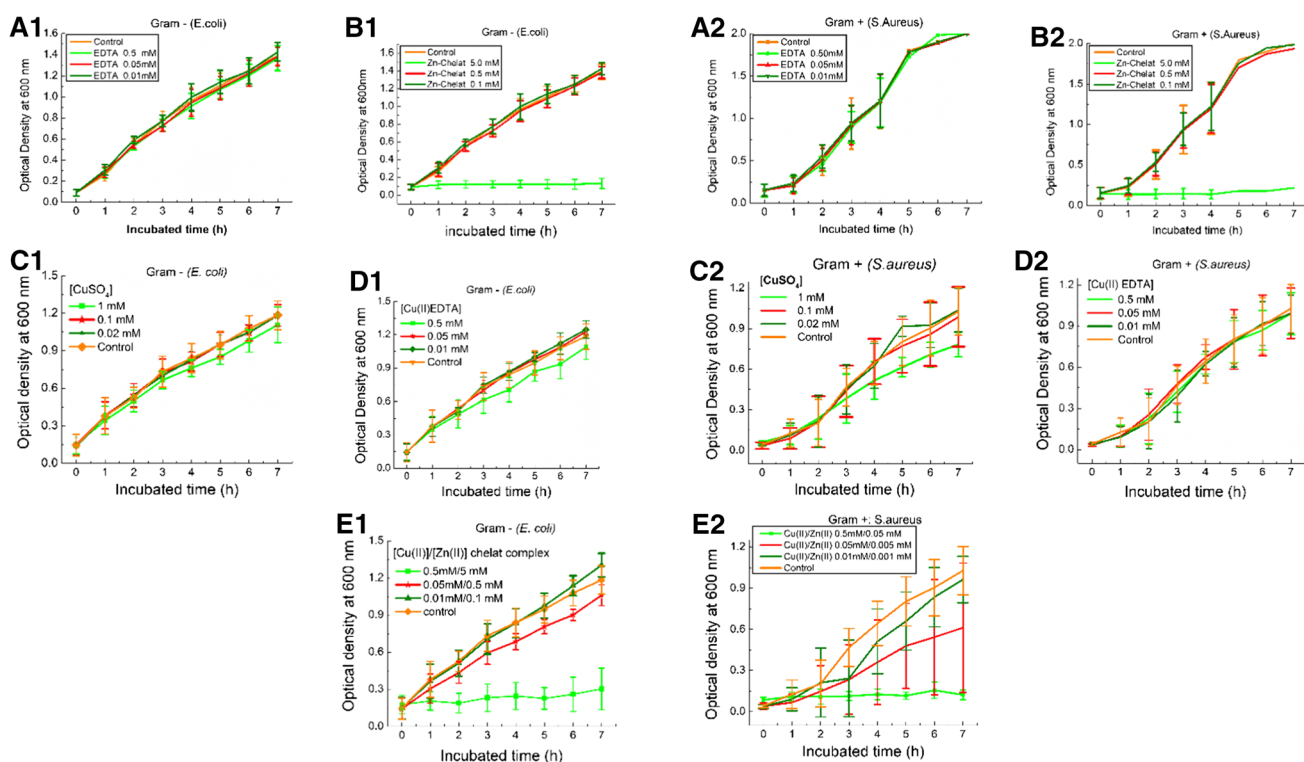


Fig. 8 Growth curves of *E. coli* and *S. aureus* in the presence of metal ions and chelating agent. A—EDTA, B—Zn²⁺ + chelate, C—CuSO₄, D—Cu²⁺ + chelate, E—Cu²⁺/Zn²⁺ + chelate; left figure: *E. coli*, right

figure: *S. aureus*. The concentrations of metal ions and chelating agents are the final concentrations in solution and are noted in the figure. Controls are bacteria without metal ion treatment

Zn²⁺ chelate concentration of 0.5 mM, there was no effect on bacterial cell growth (Fig. 8.B1, B2).

From these results, we can see that the free Cu²⁺ ion is more toxic than the Cu²⁺ chelate and Zn²⁺ chelate in the single metal ion solutions. These experiments showed

again that the combined Cu²⁺ and Zn²⁺ chelation complex effectively inhibited bacterial cell growth in both gram-negative and gram-positive bacteria (Fig. 8. E1, E2). At the highest concentration of the Cu²⁺/Zn²⁺ chelate complex, the bacterial cells did not grow at all during the 7 h

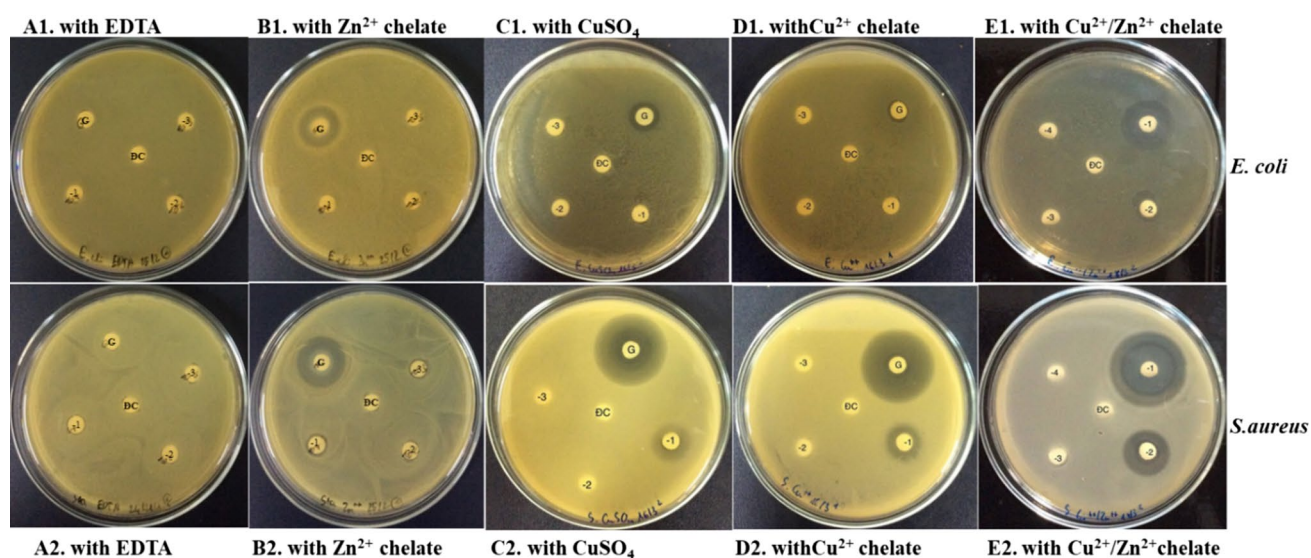


Fig. 9 Disc diffusion assay results of various types of metal ions against *E. coli* and *S. aureus*

of the experiment. However, at the lower concentration of 0.05 mM Cu^{2+} /0.5 mM Zn^{2+} chelation complex, the inhibitory effect was slightly decreased. A dose-dependent effect was observed in the case of gram-positive bacteria (Fig. 8. E2) and remained at 0.01/0.1 mM chelate complex. In the case of gram-negative bacteria (Fig. 8. E1), 0.01/0.1 mM chelate complex did not show a very different effect from the control experiment. Thus, in liquid medium, the effect of the chelate complex of Zn^{2+} and Cu^{2+} is stronger than that of Zn^{2+} or Cu^{2+} alone. The effect of metal ions on bacterial cell growth in liquid culture was evaluated as follows: $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate $>$ CuSO_4 $>$ Cu^{2+} chelate $>$ Zn^{2+} chelate. The results of these experiments were confirmed by the results obtained from the disc diffusion assay. The inhibitory effects on gram-positive bacteria were stronger than those on gram-negative bacteria. The antibacterial activity of metal ions was investigated against gram-positive (*S. aureus*) and gram-negative (*E. coli*) bacteria.

Inhibition of cell growth measured by the disc diffusion assay method

Figure 9 shows that the ZOIs of metal ions were larger in the plates loaded with *S. aureus* than in those loaded with *E. coli*. ZOIs decreased in the following sequence: $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate $>$ Zn^{2+} chelate $>$ CuSO_4 $>$ Cu^{2+} chelate $>$ EDTA in the plates loaded with *E. coli*. In *S. aureus*, the ZOIs were larger and decreased as follows: $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate $>$ CuSO_4 = Cu^{2+} chelate $>$ Zn^{2+} chelate $>$ EDTA. The results of this disc diffusion assay are also summarized in Table 2. Furthermore, at identical concentrations of metal ions, $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate had an outstanding inhibitory effect compared with the single ions or chelates, while EDTA at the same concentration used in the chelate complex had no bactericidal effect on either type of bacteria (Fig. 9. A1, A2). ZOIs of CuSO_4 at 100 mM and 10 mM (Fig. 9. C1, C2) were equal ZOIs of the Cu^{2+} chelate at 50 mM and 5 mM (Fig. 8. D1, D2) and were larger than the ZOIs of the Zn^{2+} chelate at 500 mM and 50 mM (Fig. 9. B1, B2).

From left to right: *E. coli* (1—upper figure) and *S. aureus* (2—lower figure) were treated with A- EDTA, B- Zn^{2+} chelate, C- CuSO_4 , D- Cu^{2+} chelate, and E- $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate. The metal ion concentrations used on each paper disc are detailed in Table 1.

These results can be explained by the fact that on the agar gel disc, the metal ions have a lower opportunity to bind with the amino groups than in the liquid medium, so they exist mainly as free ions on gel surfaces. These results again showed that the antibacterial activities of Cu^{2+} are higher than those of Zn^{2+} chelate. In particular, the ZOIs of the $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate at the lower concentrations of 5 mM/50 mM and 0.5 mM/5 mM (Fig. 9. E1, E2) showed stronger antibacterial activities than the single ions at the same concentrations.

Table 2 The zones of inhibition of various concentrations of metal ions

Metal ions	Symbol~Concentration (mM)	Zones of inhibition (mm)*	
		<i>E. coli</i>	<i>S. aureus</i>
$\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate	(-1)~5/50	14.30 ± 3.14	26.87 ± 1.45
	(-2)~0.5/5	8.10 ± 2.17	14.80 ± 1.91
	(-3)~0.05/0.5	6.00 ± 0.0	6.53 ± 0.67
	(-4)~0.005/0.05	6.00 ± 0.0	6.00 ± 0.0
	(DC)~DW	6.00 ± 0.0	6.00 ± 0.0
CuSO_4	(G)~100	11.97 ± 0.91	22.57 ± 0.29
	(-1)~10	6.67 ± 0.88	8.27 ± 0.51
	(-2)~1	6.00 ± 0.0	6.00 ± 0.0
	(-3)~0.1	6.00 ± 0.0	6.00 ± 0.0
	(DC)~DW	6.00 ± 0.0	6.00 ± 0.0
Cu^{2+} chelate	(G)~50	8.93 ± 0.34	25.13 ± 0.52
	(-1)~5	6.00 ± 0.0	8.80 ± 0.71
	(-2)~0.5	6.00 ± 0.0	6.00 ± 0.0
	(-3)~0.05	6.00 ± 0.0	6.00 ± 0.0
	(DC)~DW	6.00 ± 0.0	6.00 ± 0.0
Zn^{2+} chelate	(G)~500	17.98 ± 1.79	21.58 ± 0.19
	(-1)~50	7.94 ± 0.68	9.43 ± 0.37
	(-2)~5	6.00 ± 0.0	6.47 ± 0.008
	(-3)~0.5	6.00 ± 0.0	6.66 ± 0.05
	(DC)~DW	6.00 ± 0.0	6.00 ± 0.0
EDTA	(G)~50	6.00 ± 0.0	6.00 ± 0.0
	(-1)~5	6.00 ± 0.0	6.00 ± 0.0
	(-2)~0.5	6.00 ± 0.0	6.00 ± 0.0
	(-3)~0.05	6.00 ± 0.0	6.00 ± 0.0
	(DC)~DW	6.00 ± 0.0	6.00 ± 0.0

Bold values show the bacteria inhibition

* Diameter of the paper discs~6.00 mm; DC: control experiment in which paper discs were soaked in autoclaved distilled water

CuSO_4 has an inhibition effect on *S. aureus* bacteria and *E. coli* bacteria at a concentration of 10 mM. In the single metal complex, Cu^{2+} chelate has an inhibition effect on *S. aureus* bacteria but has no inhibition effect on *E. coli* bacteria at a concentration of 5 mM. Zn^{2+} chelate has an inhibition effect on both bacteria at 50 mM, while $\text{Cu}^{2+}/\text{Zn}^{2+}$ chelate still has the inhibition effect on *S. aureus* bacteria at concentration of 0.05 mM/0.5 mM and on *E. coli* bacteria at concentration of 0.5 mM/5 mM. As the result, using both Cu^{2+} and Zn^{2+} ions in the chelate complex may increase the antibacterial activity compared with the separated CuSO_4 , Cu^{2+} chelate or Zn^{2+} chelate for both bacterial types, approximately 100-fold for *S. aureus* and 20-fold for *E. coli* (Table 2).

Conclusion

These results indicated that copper ion associates more strongly than zinc ions with sulphuryl or amine groups on protein molecules. The findings also showed that gram-positive bacteria are more sensitive than gram-negative bacteria to copper ion or zinc ion solutions. In particular, the use of both Cu^{2+} and Zn^{2+} in a chelation complex with urea and EDTA enhances the antimicrobial activity against these bacteria. The bacterial inhibition by this complex was stronger than that of Zn^{2+} or Cu^{2+} alone, approximately 100-fold for *S. aureus* and 20-fold for *E. coli*. The enhanced antibacterial ability of Cu(II)/Zn(II) in chelation with EDTA and urea complex compared to their individual metals complexes has been demonstrated. In future work, the optimization of antibacterial activity by changing the ratio of the two metal ions or changing the ligand will be conducted. And the study of this complex targeting more specific bacteria such as antibiotic-resistant bacteria or bacteria in aquatic media, and exploiting their potential use for antifungals would be interesting and needed.

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